

Science, Technology and Society

Public Engagement with Science and Technology

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Today's lecture

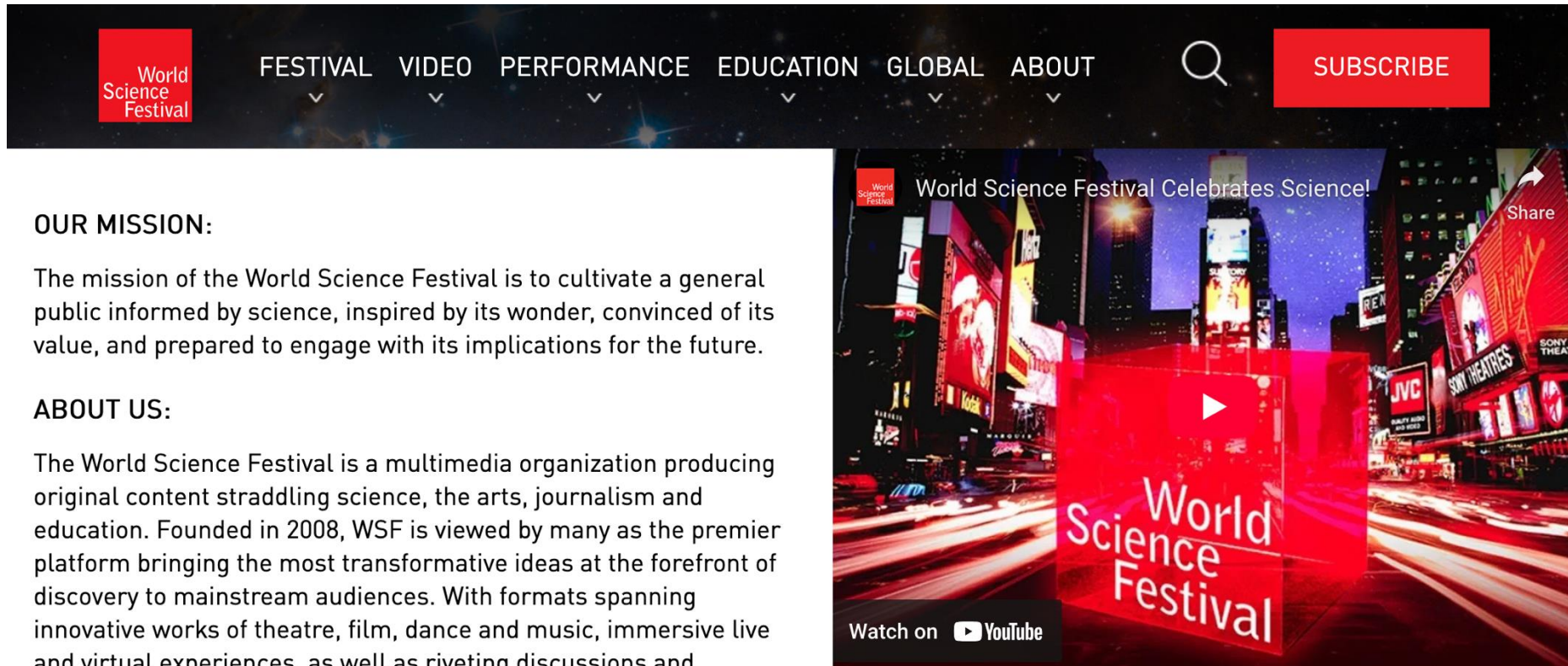


Introduction to **different approaches to technology development with stakeholders** with a focus on a) public understanding, b) public engagement, and c) citizen science.



Discussion of the **value of public participation in technological projects** from a democratic perspective and reflection on the difference between upstream and downstream engagement.

Example 1: World Science Festival



OUR MISSION:

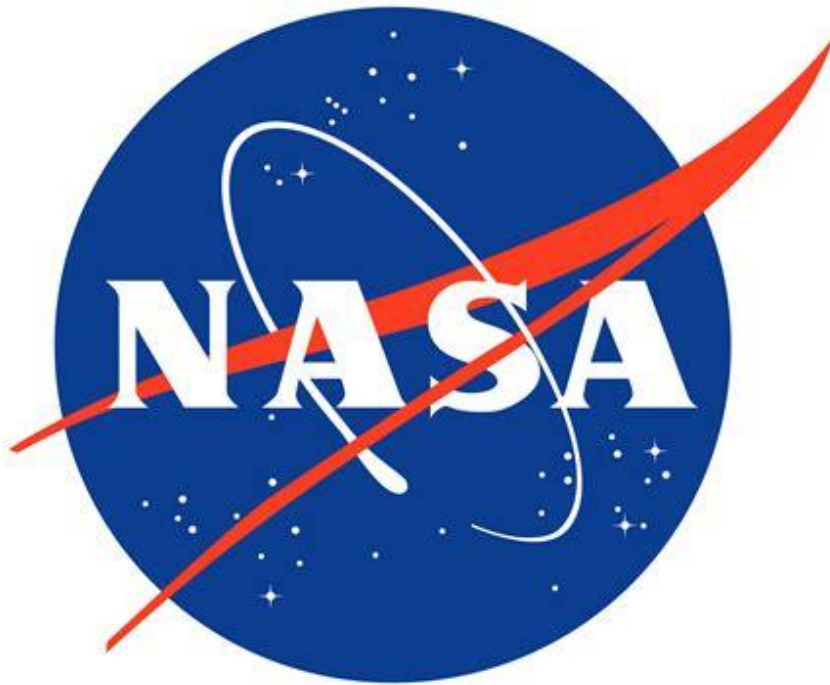
The mission of the World Science Festival is to cultivate a general public informed by science, inspired by its wonder, convinced of its value, and prepared to engage with its implications for the future.

ABOUT US:

The World Science Festival is a multimedia organization producing original content straddling science, the arts, journalism and education. Founded in 2008, WSF is viewed by many as the premier platform bringing the most transformative ideas at the forefront of discovery to mainstream audiences. With formats spanning innovative works of theatre, film, dance and music, immersive live and virtual experiences, as well as riveting discussions and debates, WSF is lauded globally for its groundbreaking live and digital content featuring world renowned scientists, artists and intellectuals, shifting science to the cultural center. Based in New York City, WSF is distinctive in its commitment to the highest of

<https://www.worldsciencefestival.com/about/>

Example 2: NASA contest



“To develop more accurate air quality data products from NASA satellite missions, a public competition called **NASA Airathon: Predict Air Quality²** asked participants to develop algorithms for estimating daily levels of surface-level air pollutants on Earth. Using NASA satellite data, model outputs, and ground measurements, the public estimated daily levels of particulate matter (PM) and nitrogen dioxide (NO₂) across urban areas in the U.S., India, and Taiwan—all of which have readily available satellite data. The contest generated more than 1,200 submissions from over 1,000 participants and awarded \$25,000 in prizes.”

Example 3: eBird (Cornell Lab of Ornithology)



Find More Birds

Explore birds and hotspots near you and wherever you go, all based on the latest sightings from around the world.



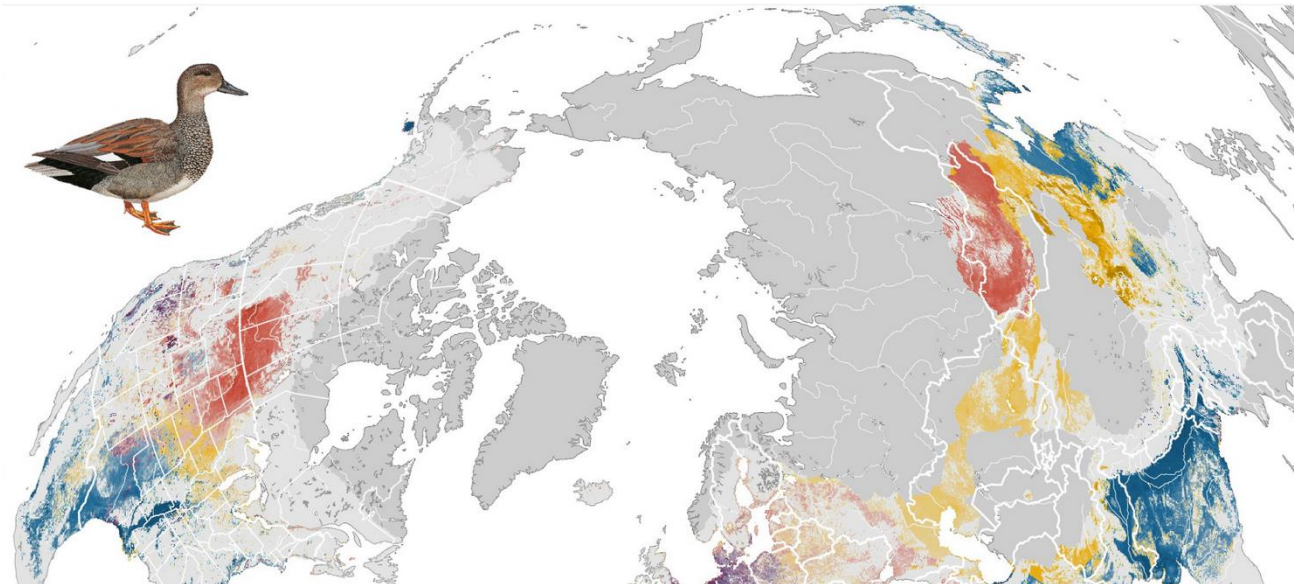
Share Your Sightings

Join the world's largest birding community. Every sighting matters. Contribute yours.



Track Your Lists

What's your latest life bird? What bird lists do you care about? eBird tallies them for you and archives your photos and sounds—all for free.



Advancing Science and Conservation

Your sightings contribute to hundreds of conservation decisions and peer-reviewed papers, thousands of student projects, and help inform bird research worldwide.

<https://ebird.org/home>

Vevox.app (ID: 136-748-124)

Which differences and which similarities do you see in how the public engages with science in the three examples?

The information deficit hypothesis

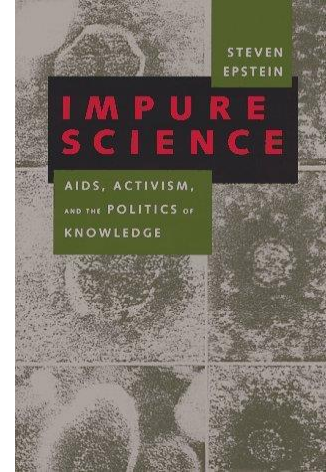
- Public acceptance of technology depends on the **public's level of literacy**.
- Lack of literacy can be mended by **communicating information from experts to the public**.



From public understanding of science to public engagement with science

1980s

Public Understanding of Science: Experts need to explain their knowledge better, i.e. in a way that makes sense to people's everyday problems.



1990s

Public Engagement with Science: The public needs to be involved in the production of knowledge, since this is where decisions are taken and courses of action enabled.

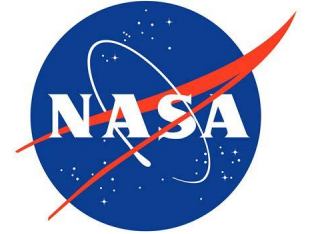
Public engagement, lay expertise and influencing research agendas



Public education model versus public consultation model



- Scientific and lay knowledge are mutually exclusive.
- Scientific knowledge draws its authority from the institution of science and depends on public trust in that institution.
- **The public must be made literate.**

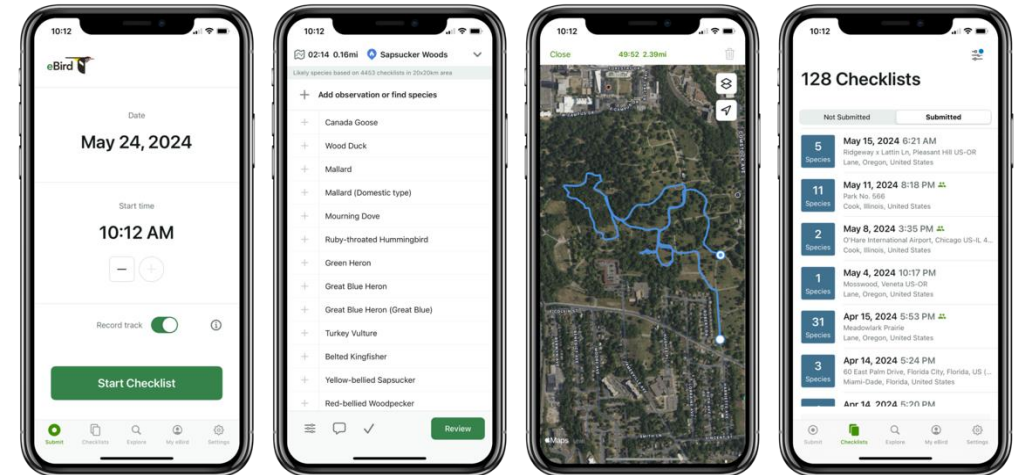


- Scientific knowledge is incomplete. Specific forms of lay knowledge can be relevant to complete the picture.
- The production of knowledge needs to address the concerns of those who are affected.
- **The public must be consulted.**

Co-creation of knowledge model

eBird

- The separation between lay and expert knowledge is to the advantage of some and disadvantage of others.
- It often only becomes evident *post facto* who profits from available expert knowledge and who do not.
- **The public must be involved.**



Some approaches for public engagement and co-creation

- Anticipatory governance (Barben et al. 2008)
- Real-time Technology Assessment (e.g., Guston and Sarewitz, 2002)
- Constructive Technology Assessment = dialogue among and early interaction with new actors (e.g., Schot and Rip, 1997)
- Value-sensitive design = approach to the design of technology that accounts for human values (Friedman 1996)
- Open innovation (Chesbrough 2023)
- Midstream modulation (Fisher et al., 2006)
- Public dialogue including consensus conferences, citizens' juries, deliberative mapping, deliberative polling and focus groups (see Chilvers, 2010)
- Etc.

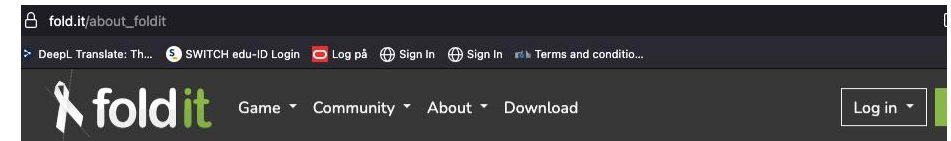
Public engagement with Science and Technology

- Public engagement (in STS) emphasises **dialogue and participation**, not one-way communication.
- **Key aims:**
 - Connect engineering decisions with **societal values and concerns**
 - Include **non-expert perspectives** in technological development
 - Increase **trust, legitimacy, and social responsibility**
- **Success indicators:**
 - Meaningful public involvement
 - Mutual learning between engineers and society
 - Public input influences real outcomes

Case Study: Foldit as a Successful Engagement Initiative

Foldit is an online game where the public helps solve **protein-folding challenges**.

- **It is successful:**
 - Engages non-experts in real scientific problem-solving
 - Combines human intuition with computational engineering
 - Public contributions led to **published scientific breakthroughs**
- **Key take away:**
- Involving society early can improve innovation quality and overall social support.



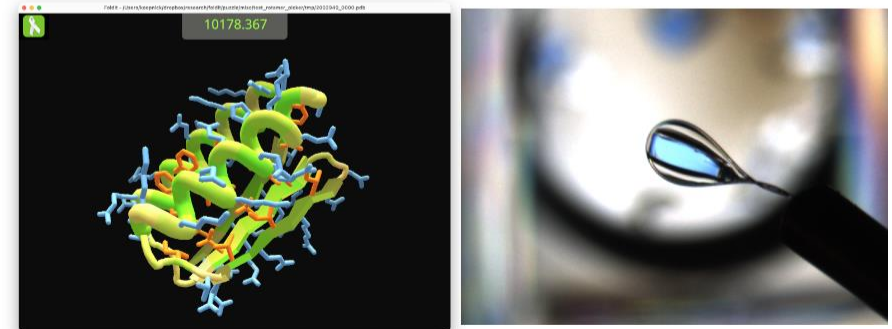
About Foldit

Foldit is a one-of-a-kind **protein folding computer game** developed by university scientists. By playing Foldit, you can contribute to advanced research on human health, cutting-edge bioengineering, and the inner workings of biology.

Foldit is **free to play** and not-for-profit. Discoveries made in the game are published in peer-reviewed research journals. Foldit players are always credited for their contributions.

Every week, Foldit scientists post new puzzles focused on the latest problems in protein folding. Read on to learn about ongoing research in **protein design** to treat diseases like influenza and COVID-19, **small molecule design** to invent new compounds, and **protein structure solving** to map the molecules that drive biology.

For a deep dive into the science behind protein folding, visit the [Science](#) page.



(Left) A protein designed by @Waya, @Galaxie, and @Susume and (right) a crystal of this protein in the lab, ready for X-ray diffraction. This protein design was published in Koepnick et al. (2019) as Foldit1, and the solved structure is deposited in the PDB with ID 6MRR.

Vevox.app (ID: 136-748-124)

**How would you describe
public engagement in the
Energy Island debate?**

Activity - Vevox.app (ID: 136-748-124)

- Search and choose a recent example of a public engagement initiative related to science. Possible examples could be a science festival, a popular science YouTube channel, a government-initiated public consultation on scientific issues, or a citizen science project. For inspiration you can look at relevant academic journals (e.g., Public Understanding of Science), news or social media reportage, or conduct an internet search of relevant initiatives (e.g., Zooniverse).
- **Analyse the public engagement initiative by discussing a) who initiated it for whom and why?, b) how effective was the initiative and how would you assess this?, c) what limitations of the initiative could you find (e.g., accessibility)?**

DTU



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